Mid Semester Examination 2025

Name of the Program: **B.Tech. (Electrical Engineering)** *Semester: 4th*
Name of the Course: Mathematics – III
Course Code: TMA 401

Time: 90 min Maximum Marks: 50

Note:

(i) Answer **all the questions** by choosing ***any one of the sub questions***.
(ii) Each question carries 10 marks.

Q1	(10 marks)	
(a)	Suppose a student dormitory in a college consists of: 30% freshmen of whom 10% own a car; 40% sophomores of whom 20% own a car; 20% juniors of whom 40% own a car; 10% seniors of whom 60% own a car; Find the probability that a student in the dormitory own a car.	CO1
	OR	
(b)	If 12 persons are seated at a round table, what is the probability that two particular persons sit together? If 6 boys and 6 girls sit in a row randomly, find the probability that all the 6 girls sit together.	
Q2	(10 marks)	
(a)	Three machines, X, Y, Z produce respectively 50%, 30% , 20% of the total number of item of factory. The percentages of faulty output of this machine are respectively 3%, 2%, 4%. An item is selected at random from the product and is found faulty. Find the probability that item was manufactured by machine B.	CO1
	OR	
(b)	The mean and variance of a binomial variate X with parameters n and p are 16 and 8. Find the following: (i) $P(X=0)$ (ii) $P(X=1)$ (iii) $P(X \geq 2)$.	
Q3	(10 marks)	
(a)	The diameter of an water pipe, say X, is assumed to be a continuous random variable with *probability density function* $f(x) = 6(x - x^2), 0 \leq x \leq 1$. Determine $\boldsymbol{\alpha}$ such that $\boldsymbol{P(X < \alpha) = P(X > \alpha)}$.	CO2
	OR	
(b)	The random variable X has a Poisson distribution. If $P(X=1) = 0.01487$, $P(X=2) = 0.04461$. Then find $P(X=3)$.	
Q4	(10 marks)	
(a)	Let X be a continuous random variable with probability density function $f(x) = \begin{cases} k\,x & ;0 \leq x \leq 1 \\ k & ;1 \leq x \leq 2 \\ -kx + 3k & ;2 \leq x \leq 3 \\ 0 & ;elsewhere \end{cases}$ (i)Determine the value of k (ii) Mean and variance of X	CO2
	OR	

(b)	A die is tossed thrice. A success is getting 1 or 6 on a toss. Find the mean and the variance of the number of successes.	
Q5	(10 marks)	**CO1**
(a)	If X is random variable for probability mass function is $p(x) = \begin{cases} \frac{x}{15}, & x = 1,2,3,\ldots. \\ 0, & elsewhere \end{cases}$ Find (i)$P(X = 1 or 2)$ and (ii) $P\left(\frac{\frac{1}{2}<X<\frac{5}{2}}{X>1}\right)$	
	OR	
(b)	A and B play a game in which their chances of winning are in the ration 3:2 . Find A's chance of winning at least three games out of the five game played.	